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“Security is a real need in a smart grid, which is currently not adequately addressed”

Smart grid technology blends communication technologies with the power grid to create a whole new smarter grid that intelligently improves efficiency and reliability.

David Andeen, segment manager, Smart Grid, Maxim Integrated, an American company that designs, manufactures and sells analogue and mixed-signal semiconductor products, spoke to Dilin Anand and Ashwin Gopinath of EFY about the smart grid and how to design robust next-gen solutions for it

What are the main areas of focus for the smart grid, and why?

The primary focus is on building more functional, integrated smart meters. Smart meters are the volumes and revenue driver in smart grid growth right now, which, in turn, drives the innovation. We envision a single chip providing metrology, security and communication. We are already seeing development of a family of products.

What are the hardware design challenges while integrating such a multitude of functionalities?

This question drives to the heart of analogue integration. Integrating multiple analogue functions into one chip is a tremendously difficult task because it involves merging functions that were originally optimised in different process geometries. Ultimately, the biggest challenge is in making the performance of analogue functions similar or better. In smart meter chips, metrology is an analogue function where we accurately detect energy across a wide current range.

What does a smart grid solution need to become competitive?

Smart grid solutions need to provide the highest-accuracy metrology or measurement of the energy. We do this in electricity meters and in distribution automation equipment. We combine the highest accuracy with a high level of analogue and digital integration to produce metrology SoCs.

The next generation of SoCs will feature security and communication. Security is a real need in a smart grid, which is currently not adequately addressed. Metrology and security make these products the most competitive.

What communication technologies will be implemented into metrology SoCs?

The current focus is on powerline communication, where we have defined G3 and we are also implementing additional solutions. We have also partnered with RadioPulse for ZigBee solutions. In the future, we will implement additional protocols. We see integration as the primary path for all communication protocols.

For design engineers planning to implement a metrology SoC in their products, what do you suggest to shorten the time to market?

Design engineers have a tough job. In addition to hardware, smart meters require a lot of firmware and software for complete operation. These devices have gone from being high-performance meters to being network nodes that still require high-performance metrology.

Design engineers need to convince themselves that they are implementing the best possible solution. They need to read application notes and firmware documents to be sure that they are completely familiar with the system.

It is difficult to give advice on shortening a process that requires a lot of time and effort. The best way to save time is to make good decisions. More time spent making a decision is almost always less than the time required to change a decision.

What are the customer demands?

Great question, because smart grid is not just about supplying parts, it is about providing solutions. There are nu-